Design and Characterization of a 32 GHz Differential Traveling-Wave Mach-Zehnder Modulator for Radio-Over-Fiber Application

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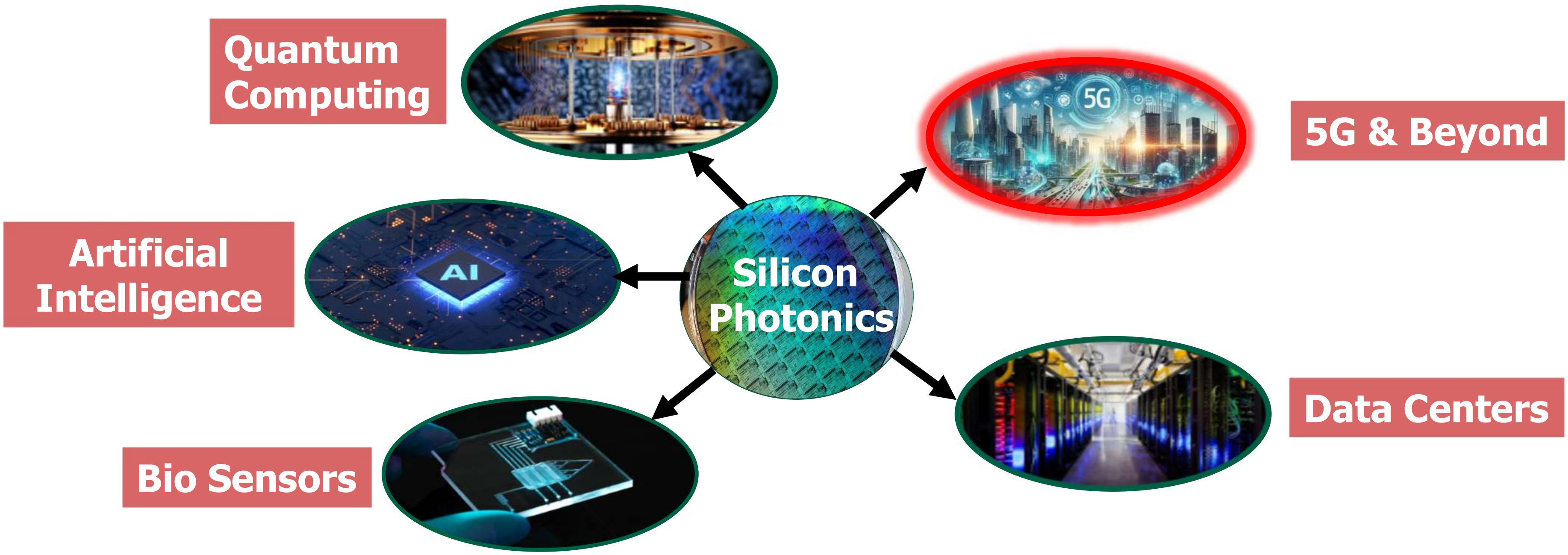
Outline

- Motivation
- Mach-Zehnder Modulator
- Differential Drive Silicon Photonic Mach-Zehnder Modulator Design and Characterization
- A 16-32 GHz Radio-Over-Fiber Integrated Front-end Implementation
- Conclusion

Photonics – Applications



- Next-gen 5G and beyond benefits from Radio-Over-Fiber (RoF) architecture to meet congestion demands and provide network access to remote locations

Analog Radio-Over-Fiber - Overview

- 5G and beyond demand low power, low latency, and high-speed
- Radio-Over-Fiber (RoF) is implemented to centralize baseband processing, thus reducing load on remote units
- Digital RoF is evolved to support low latency but demands power hungry analog and digital components with increasing speed and bandwidth requirements
- Analog RoF supports mm-wave 5G by completely transferring baseband processing and high-power components to central location

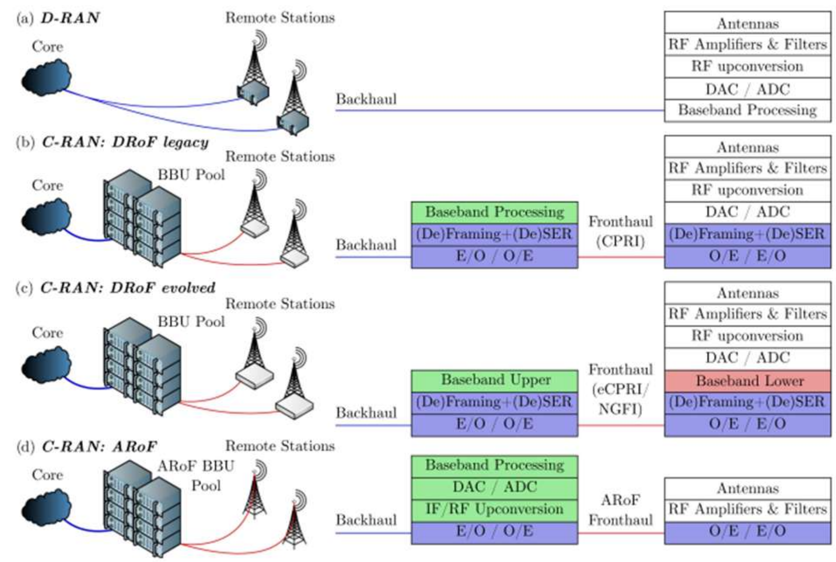


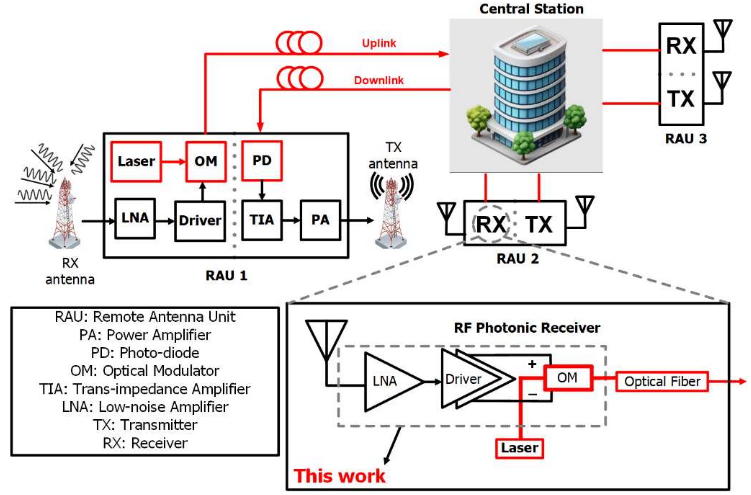
Fig. 1. Comparison of RAN options and corresponding placement of signal processing functions: (a) D-RAN with full baseband processing at the remote site, (b) traditional C-RAN with DRoF CPRI fronthaul and centralized baseband processing, (c) evolved C-RAN with DRoF eCPRI or NGFI fronthaul and functional split, (d) C-RAN with ARoF fronthaul and full centralization of baseband processing and RF sources. SER: seralization, O/E & E/O: optical to electrical and electrical to optical conversion.

S. Rommel et al., "Towards a Scaleable 5G Fronthaul: Analog Radio-over-Fiber and Space Division Multiplexing," in Journal of Lightwave Technology, vol. 38, no. 19, pp. 5412-5422, 1 Oct.1, 2020, doi: 10.1109/JLT.2020.3004416.

Analog Radio-Over-Fiber - Architecture

- Multiple remote antenna units (RAUs) are deployed to improve coverage
- High-speed optical modulator is the key component in the receiver front-end
- Silicon photonic modulator can be potentially integrated with CMOS process
- This work focusses on silicon photonic Mach-Zehnder modulator design that supports integration with CMOS driver to demonstrate a RF photonic receiver front-end

Radio-Over-Fiber



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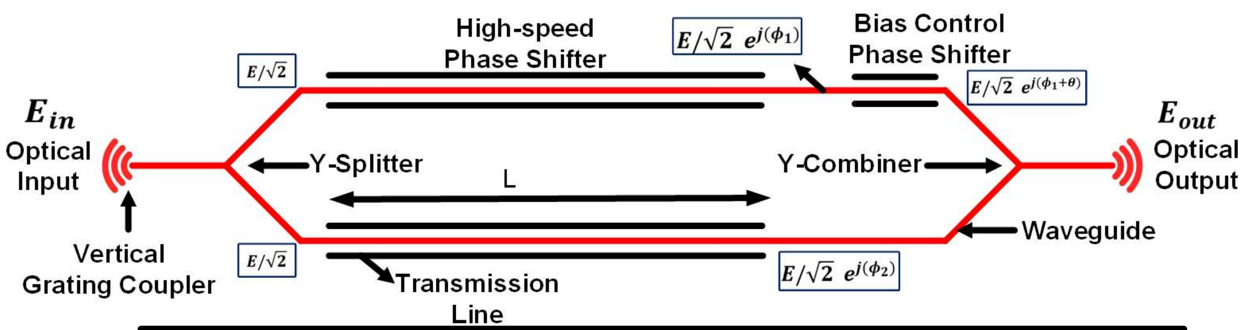
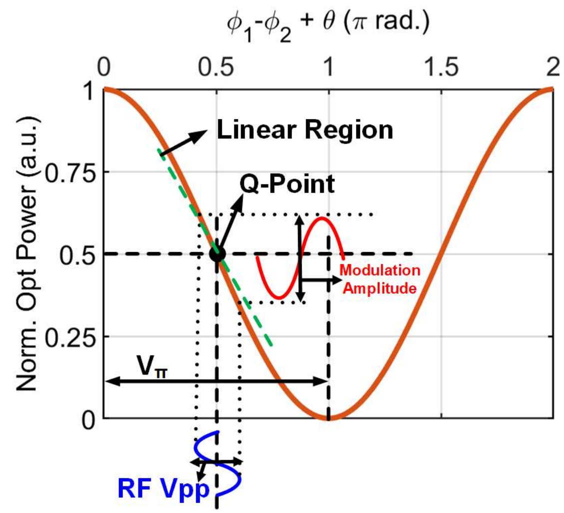
Mach-Zehnder Modulator (MZM) – Working Principle

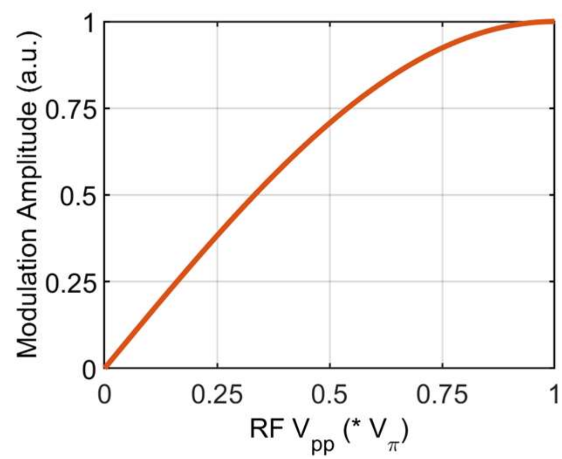


$$P_{in} = |E_{in}|^2$$

$$E_{out} = \frac{E_{in}}{2}\left(e^{j(\phi_1+\theta)} + e^{j\phi_2}\right)$$

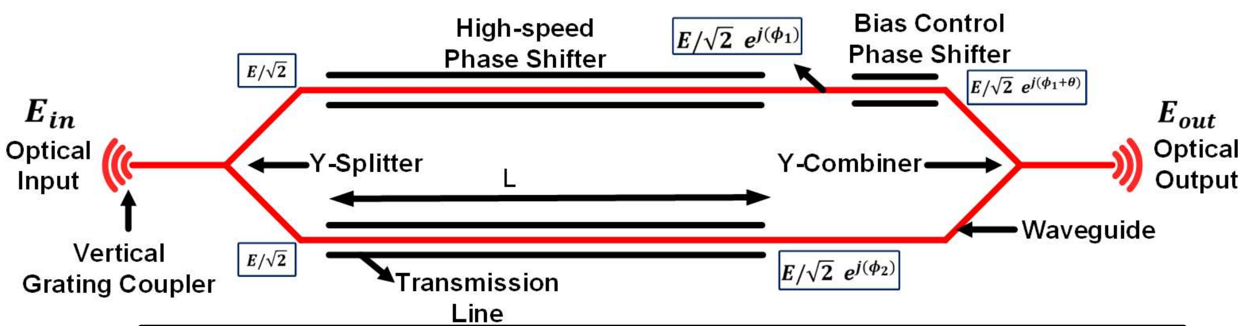
$$P_{out} = |E_{out}|^2 = \frac{P_{in}}{2}\left(1 + cos(\phi_1 - \phi_2 + \theta)\right)$$

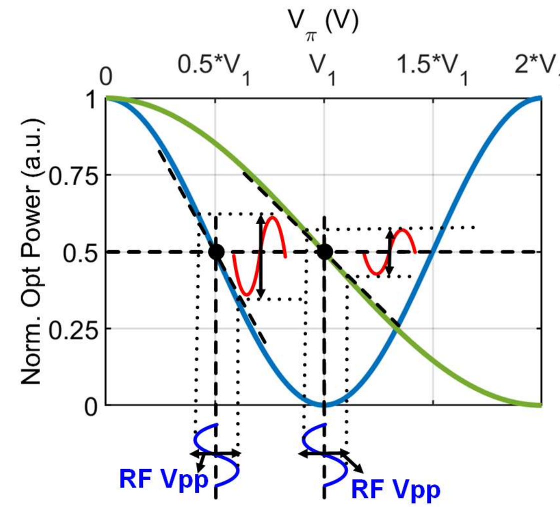




- Works on the principle of constructive and destructive interference
- Operated at Q-point along the linear region for intensity modulated optical signal of RF signal
- V_π is the voltage required for the MZM output power to switch from peak to minimum defined by the modulation efficiency $V_\pi * L\ (V - cm)$

Mach-Zehnder Modulator (MZM) – Design Metrics





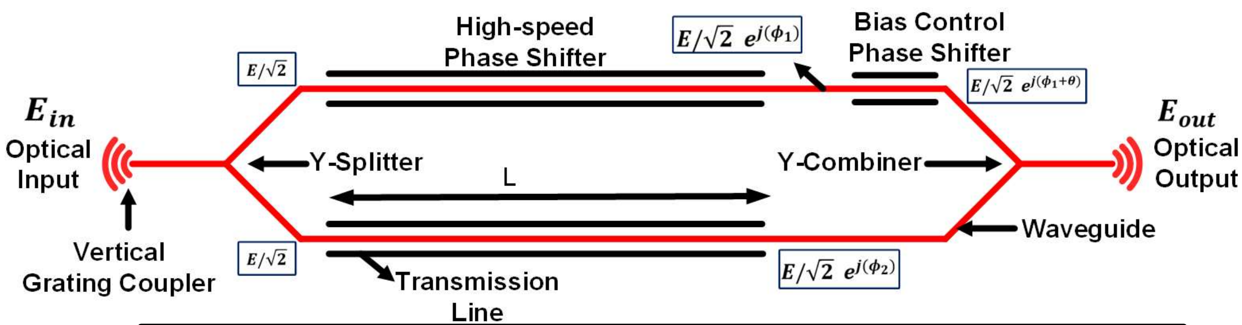
$$P_{in} = E_{in}^2$$

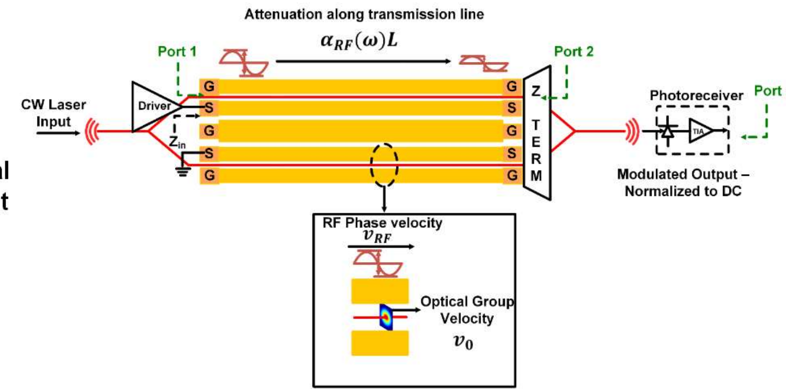
$$E_{out} = \frac{E_{in}}{2}\left(e^{j(\phi_1+\theta)}+e^{j\phi_2}\right)$$

$$P_{out} = E_{out}^2 = \frac{P_{in}}{2}\left(1 + cos(\phi_1 - \phi_2 + \theta)\right)$$

- Higher V_π translates to higher linearity (slope at Q-point) but lowers modulation amplitude impacting front-end signal-noise ratio
- Longer MZM improves V_π but RF loss of transmission line limits the MZM electric – electric 3-dB bandwidth

Mach-Zehnder Modulator (MZM) – Design Metrics





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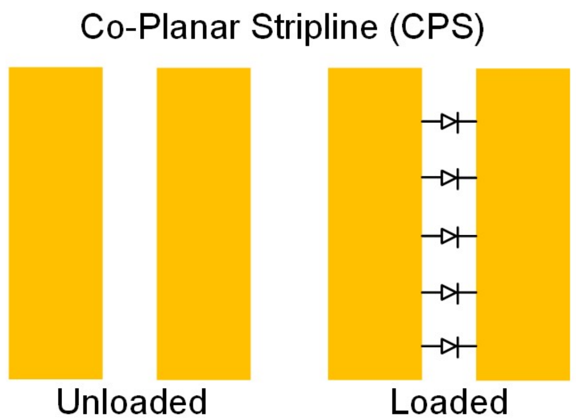
$$P_{out} = E_{out}^2 = \frac{P_{in}}{2}\left(1 + cos(\phi_1 - \phi_2 + \theta)\right)$$

Rule-of-thumb $(v_{RF} = v_0, Z_{in} = Z_0 = Z_{term})$

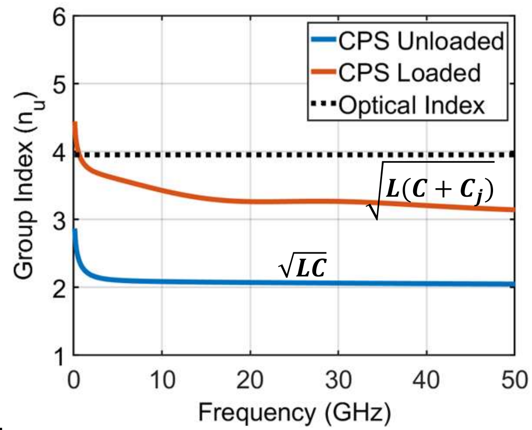
$$3dB\ EE\ S31 \approx 6.4dB\ TL\ S21$$

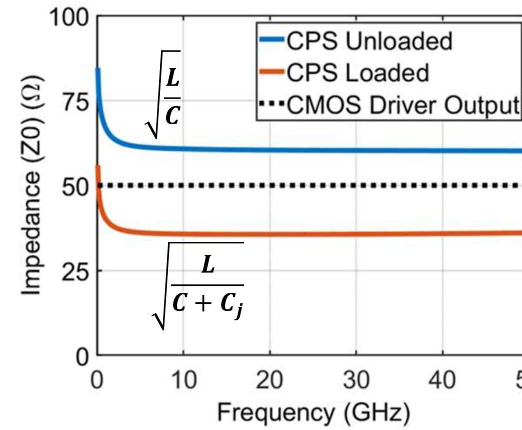
- Electric-Electric 3-dB bandwidth includes the effect of transmission line loss, RF-optical velocity matching, and impedance matching at source and termination
- Under perfect RF-optical velocity matching and impedance matching at source and termination of the transmission line the 6.4 dB transmission line bandwidth is approximately equal to the MZM 3-dB electric-electric bandwidth

Silicon Photonic MZM for CMOS Integration – Design Challenges



Diode represents the PN junction of high-speed phase shifter





L, C – Unit inductance (H/m) and capacitance (F/m) of transmission line
C_j - PN junction voltage dependent capacitance

- A simple co-planar strip line does not provide required RF group index and impedance
- Slow-wave electrode design is required to match RF and optical velocities along with impedance matching at CMOS-PIC interface

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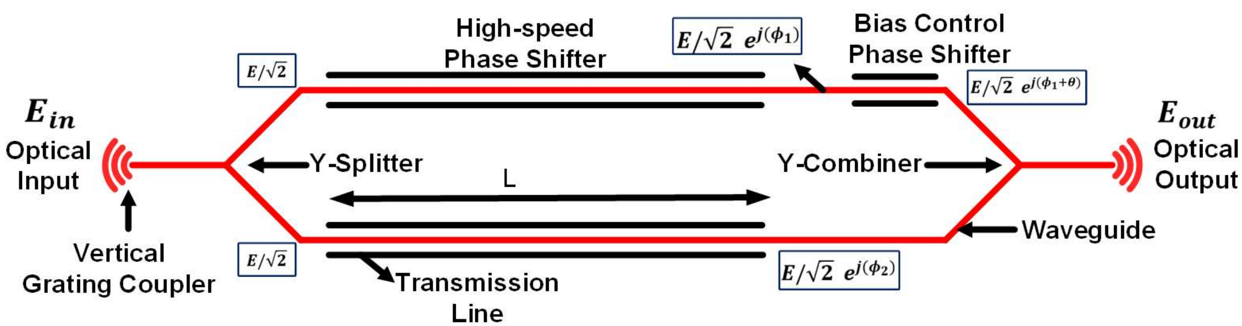
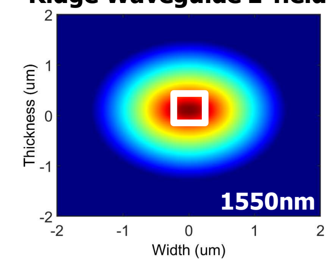
Mach-Zehnder Modulator (MZM) – Components

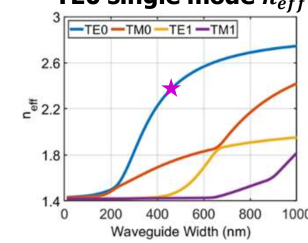


- Grating couplers are used for coupling 1550nm continuous wave laser in and out of the MZM
- Silicon waveguides with 500nm x 220nm core region transmit the TE0 mode of coupled laser along the MZM
- A low-loss Y-junction performs splitting and combination of light at start and end of phase shifter arms respectively
- Change in effective index of optical wave in silicon waveguide is achieved by the plasma dispersion effect (high-speed) or thermal effect (low-speed)

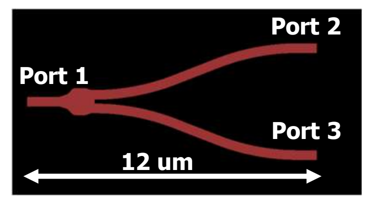
Ridge Waveguide E-field

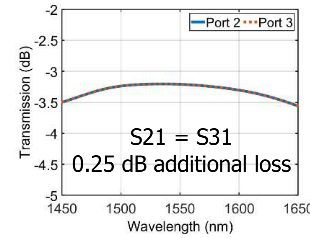


TE0 single mode n_{eff}

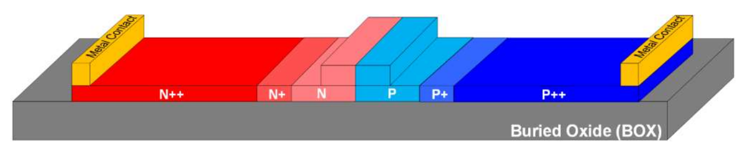


Compact and low-loss Y-junction[1]

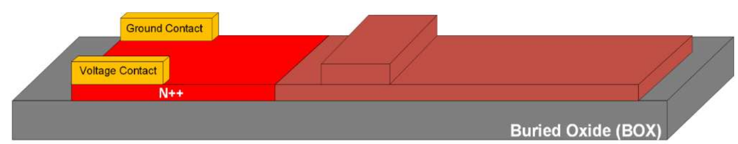




High-Speed PN Depletion Phase Shifter



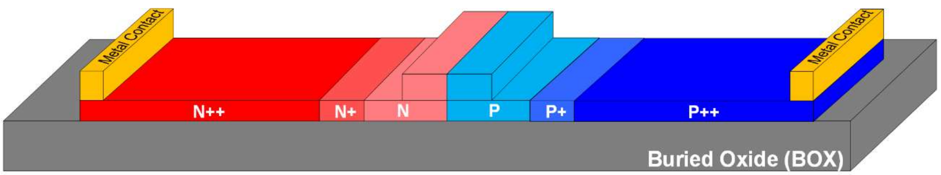
Low-Speed Thermal Phase Shifter



[1] Yi Zhang, Shuyu Yang, Andy Eu-Jin Lim, Guo-Qiang Lo, Christophe Galland, Tom Baehr-Jones, and Michael Hochberg, "A compact and low loss Y-junction for submicron silicon waveguide," Opt. Express 21, 1310-1316 (2013)

Silicon Photonic Phase Shifter

High-Speed PN Depletion Phase Shifter



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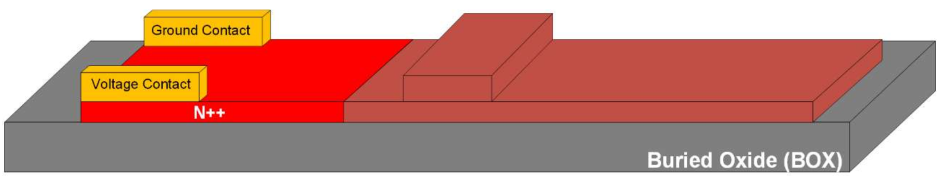
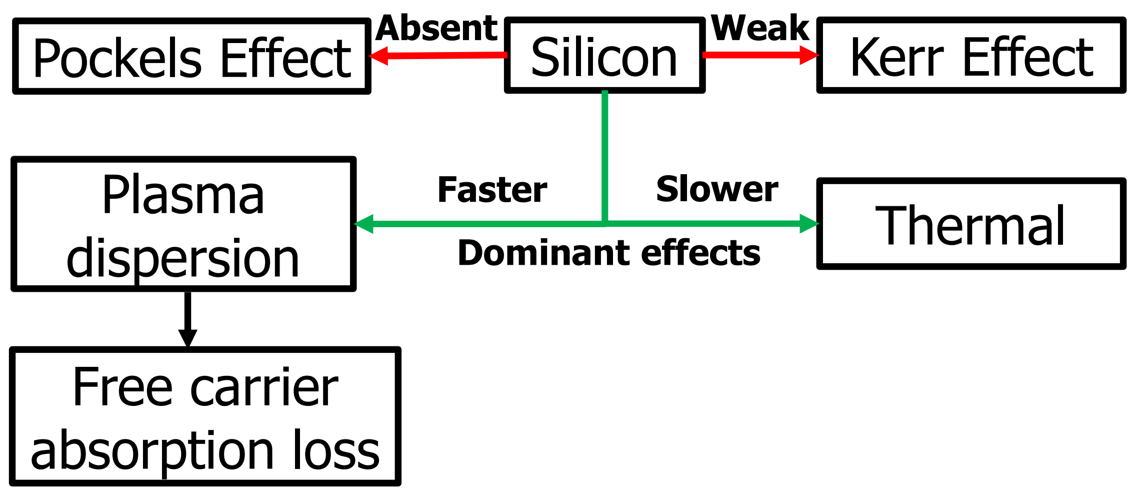
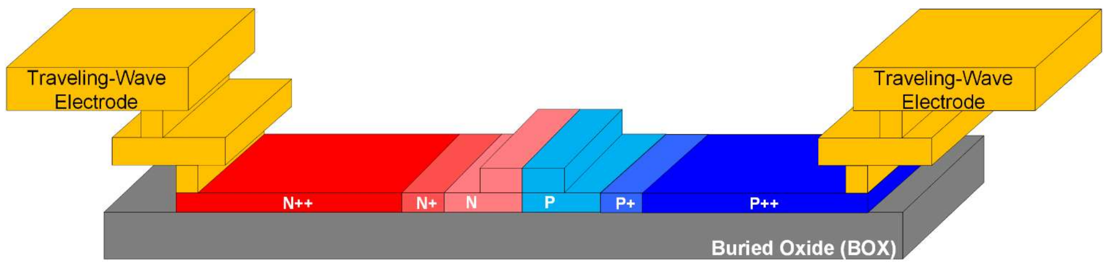


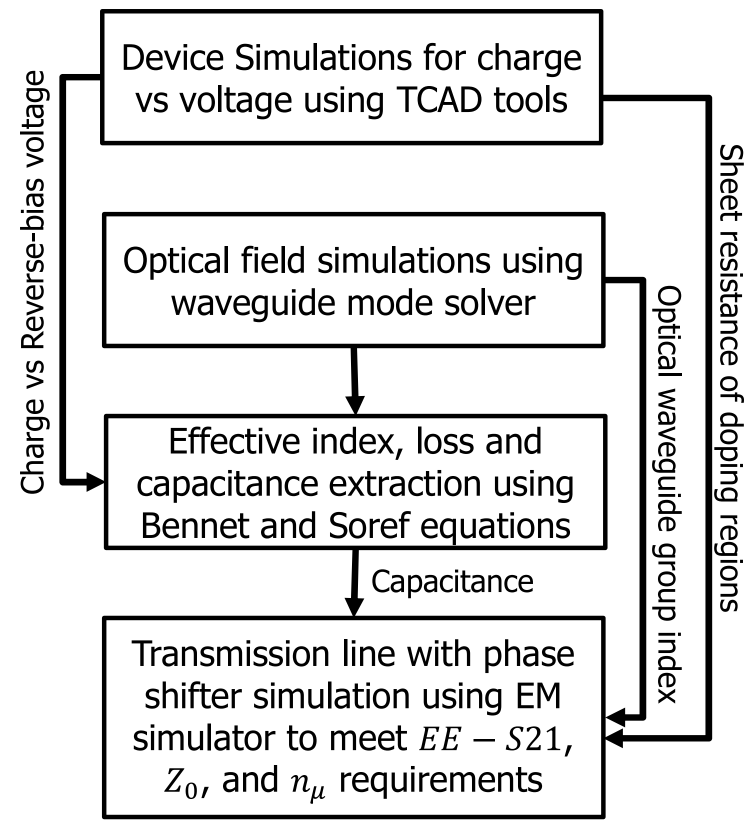


- PN depletion mode phase shifter enables high-speed performance by applying the voltage swing with a DC reverse bias voltage
- Phase shift vs voltage is non-linear due to non-linearity from PN junction capacitance
- Low-speed bias control phase shifter is implemented using lateral thermal phase shifter with N++ doped poly resistor

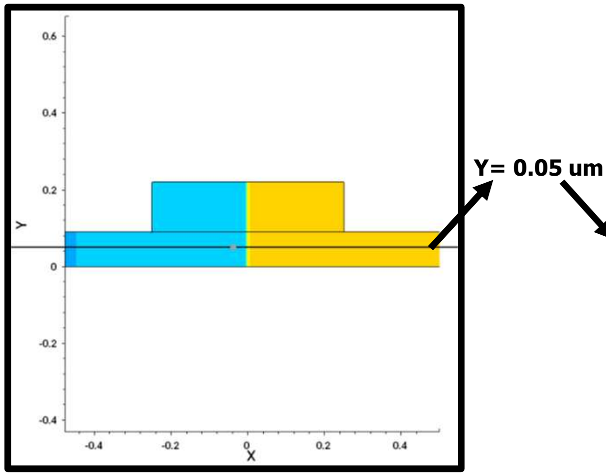
MZM – High-Speed Phase Shifter Design

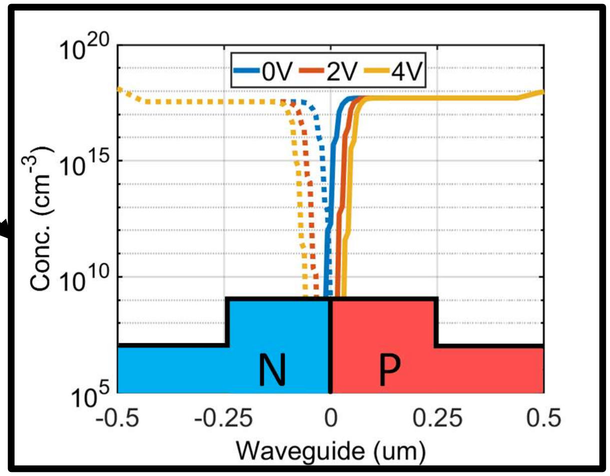


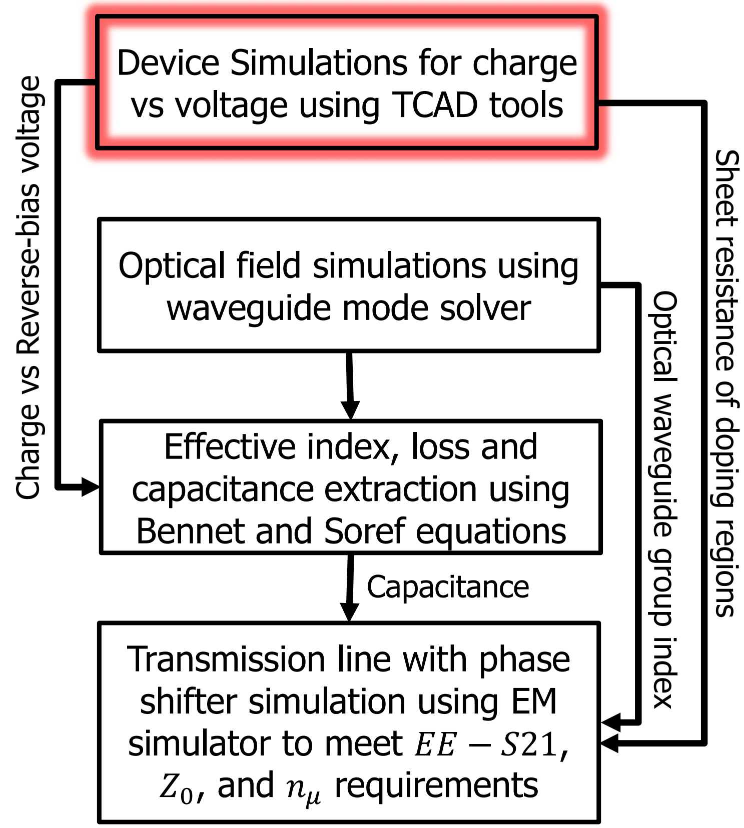
- Waveguide device and optical simulations are performed to change in effective index (Δn_{eff}), loss $\Delta\alpha\ (dB/cm)$ PN junction capacitance$\left(C_j\ fF/um\right)$
- Traveling-wave electrode with phase shifter defined using sheet resistance and junction capacitance is simulated to determine the high-speed performance of the MZM



MZM – High-Speed Phase Shifter Design

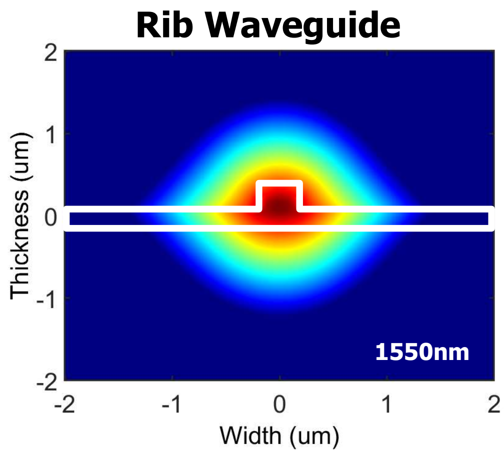






- A silicon waveguide with 3-level doping is defined based on doping concentration calculated from the sheet resistance values provided by the foundry [https://www.pvlighthouse.com.au/resistivity]
- 2-D charge profile for different reverse bias voltages is extracted using Synopsys TCAD tools

MZM – High-Speed Phase Shifter Design



2D - Optical Field

$E(x,y)$ @ $1550nm$

2D – Charge Distribution

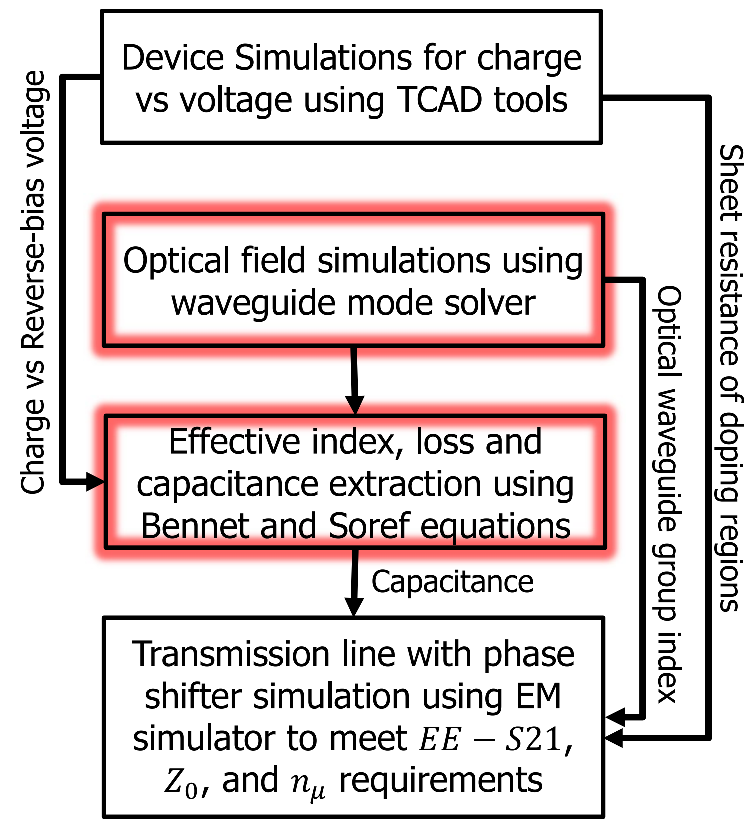
$\Delta e(x,y,V), \Delta h(x,y,V)$

$$\Delta n(x,y,V) = -8.8e^{-22} * \Delta e(x,y,V) - 8.5e^{-18} * (\Delta h(x,y,V))^{0.8}$$

$$\Delta \alpha(x,y,V) = 8.5e^{-18} * \Delta e(x,y,V) + 6e^{-18} * \Delta h(x,y,V)$$

$$\Delta n_{eff}(V) = \int\int \frac{(E^*(x,y) * \Delta n(x,y,V) * E(x,y)dxdy)}{E^*(x,y) * E(x,y)dxdy}$$

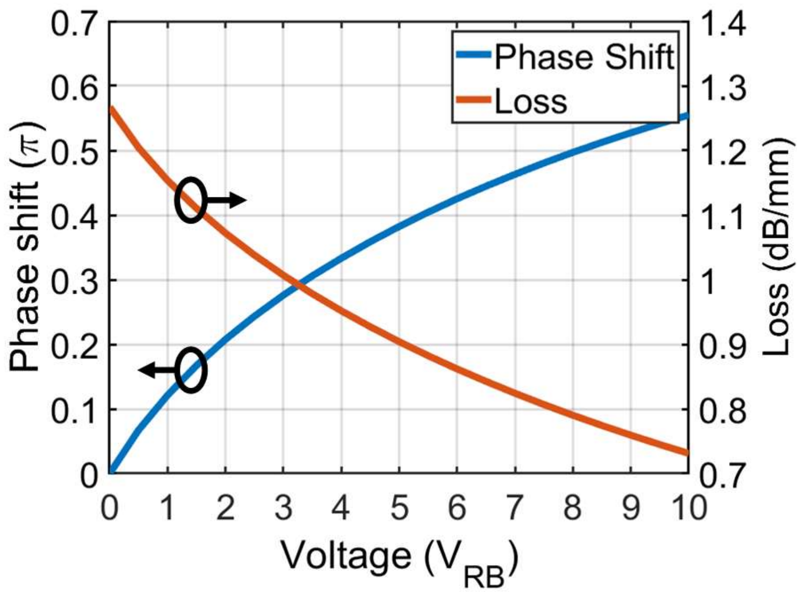
$$\Delta \alpha(V)\ (cm^{-1}) = \int\int \frac{(E^*(x,y) * \Delta \alpha(x,y,V) * E(x,y)dxdy)}{E^*(x,y) * E(x,y)dxdy}$$

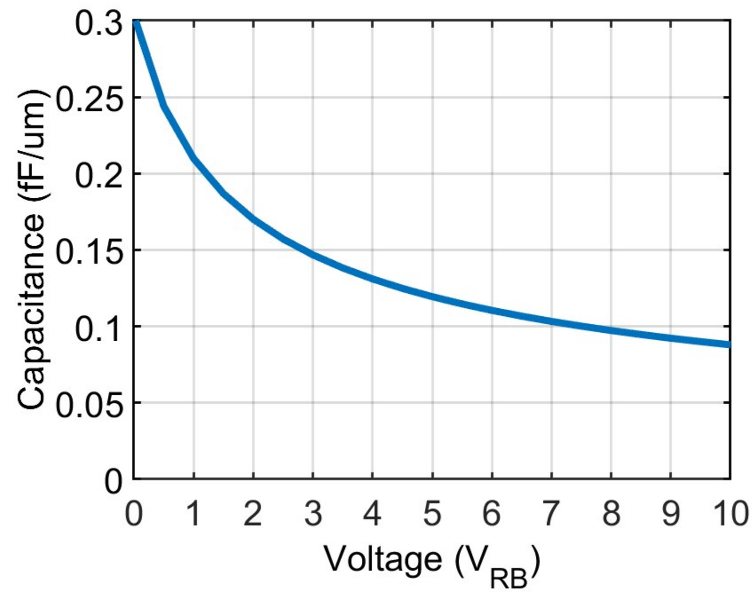


- 2D optical field at 1550nm is simulated using Lumerical waveguide mode solver
- Using Bennet and Soref equations [2] the simulated charge profile and optical field are integrated to calculate the Δn_{eff}, loss $\Delta\alpha$, and C_j

[2] R. Soref and B. Bennett, "Electrooptical effects in silicon," in IEEE Journal of Quantum Electronics, vol. 23, no. 1, pp. 123-129, January 1987, doi: 10.1109/JQE.1987.1073206.

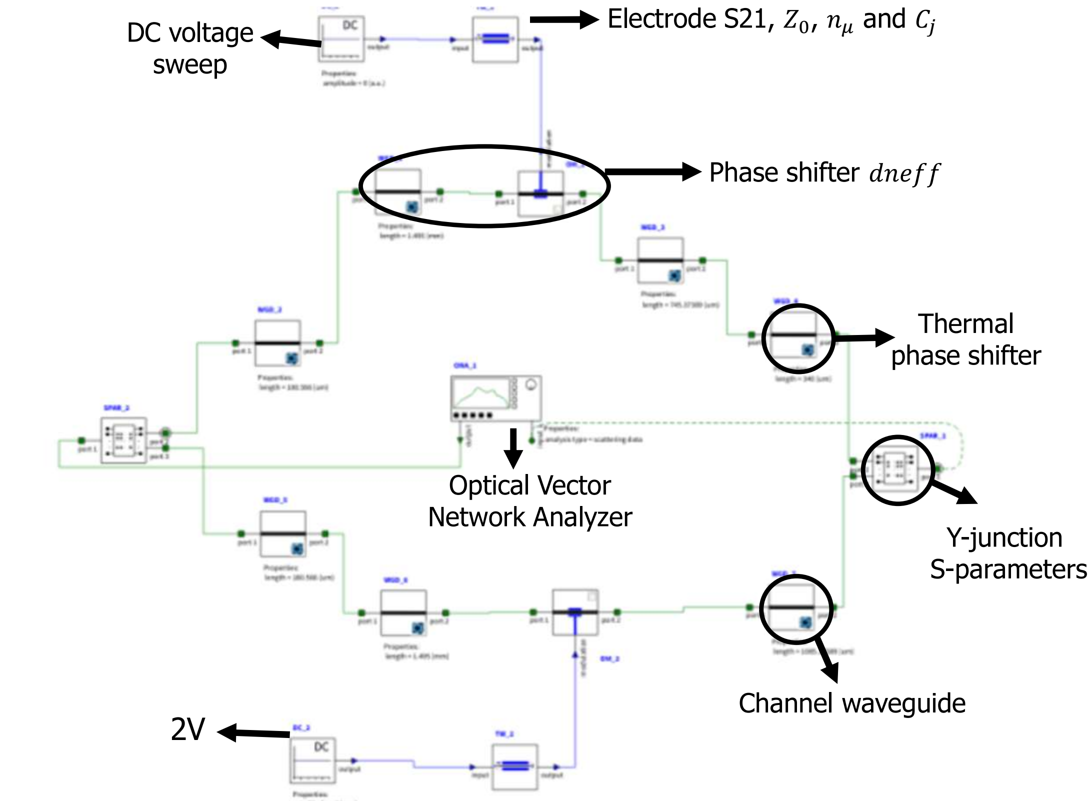
MZM – Phase Shift Performance

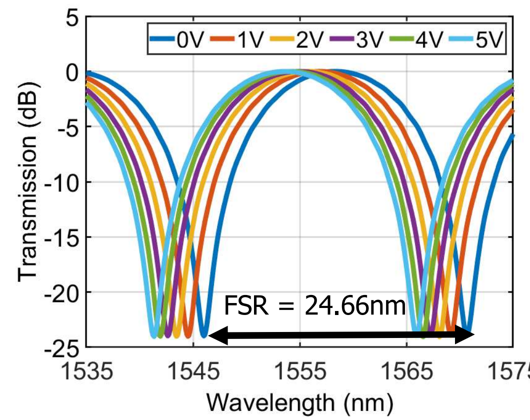


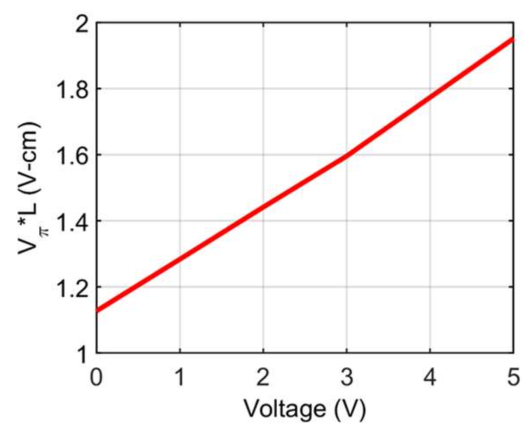


- $V_{\pi} * L = 1.45\ V - cm$ and Loss = 1.15 dB/mm @ 2V
- Effective length of phase shifter is 1.35mm

MZM – Optical Performance



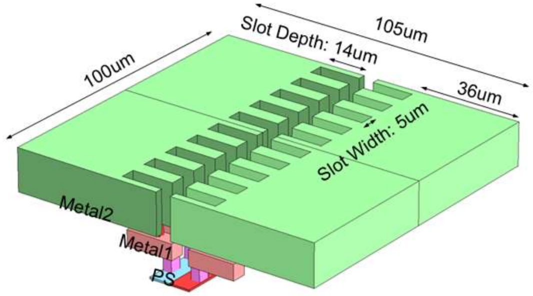




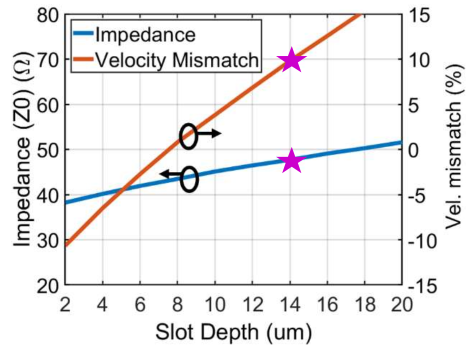
- $\Delta\phi = 2\pi * \frac{\Delta\lambda(V)}{FSR},\ V_\pi * L = \frac{V*\pi}{\Delta\phi} * L = 1.45\ V - cm\ @2V$

MZM – High-Speed Phase Shifter Design

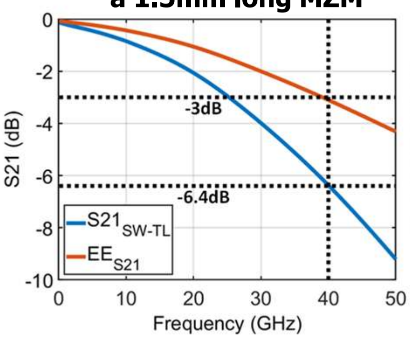
Slow-wave transmission line unit cell with inductive slots



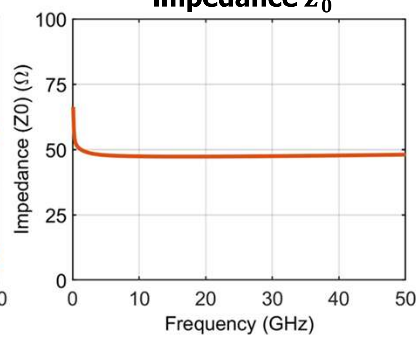
Inductive slot depth optimization



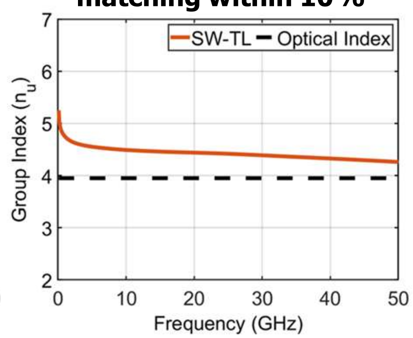
38.8 GHz EE-S21 bandwidth of a 1.5mm long MZM



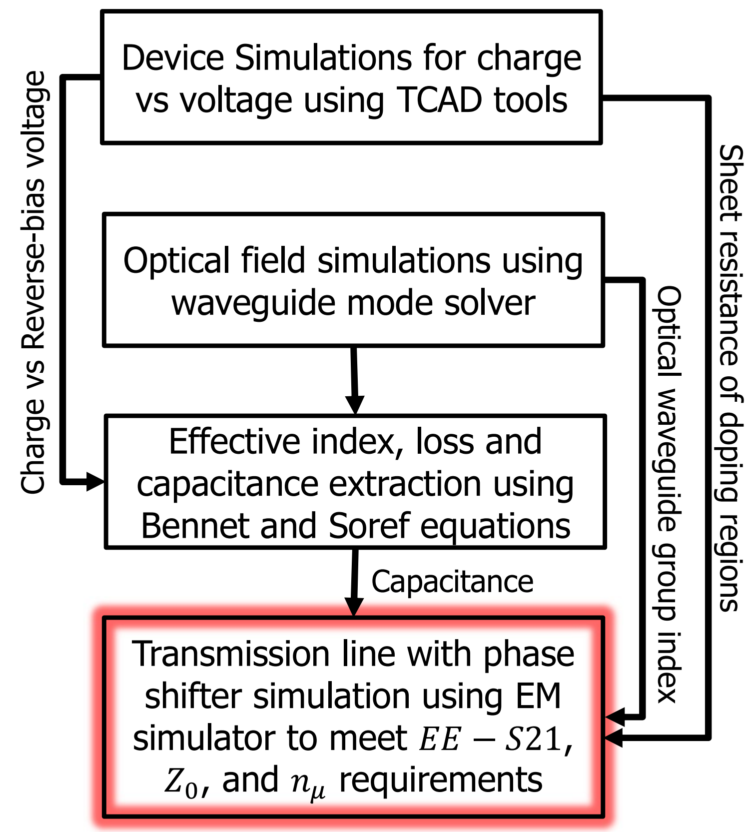
47.5 Ω transmission line impedance Z_0



RF-Optical velocity matching within 10%

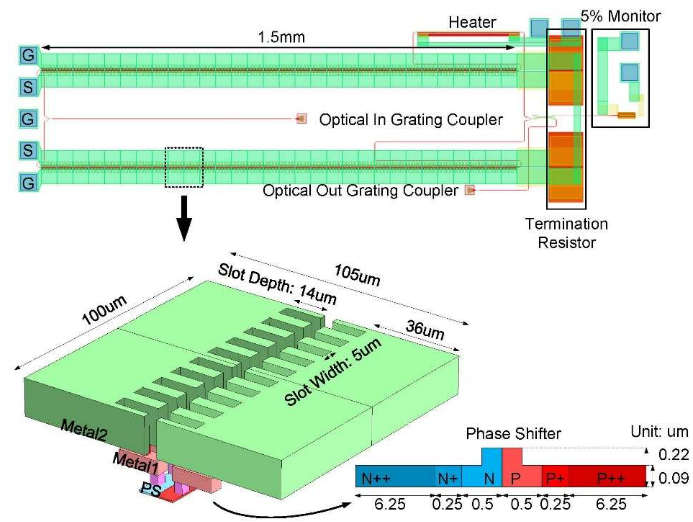


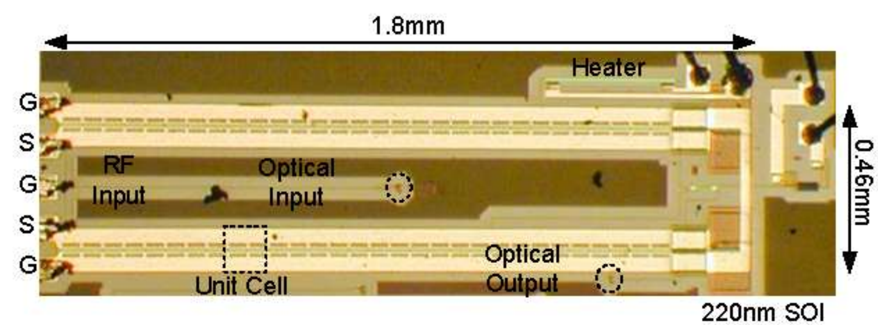
- Inductive slots help achieve simultaneous RF-optical velocity matching and closer to 50Ω Z_0 for efficient CMOS integration



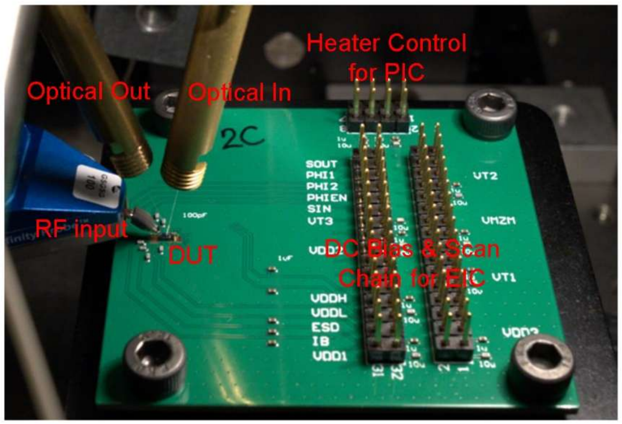
MZM – Layout and Micrograph

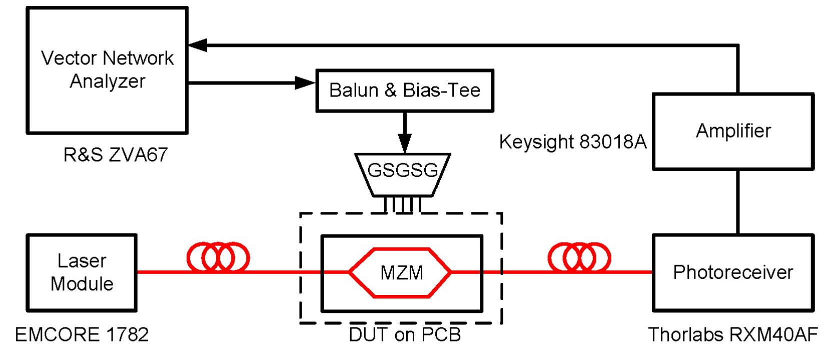
- Measurement ready layout with strategic placement of vertical fiber grating couplers and RF GSGSG pads enable standalone MZM performance characterization and seamless CMOS integration
- Phase shifter uses three level doping to reduce free carrier absorption loss
- Waveguides are routed at least 100um away from heater to reduce thermal cross-talk
- Fabricated MZM is mounted on a FR4 PCB
- Voltage to heater is supplied by bondwire from PCB

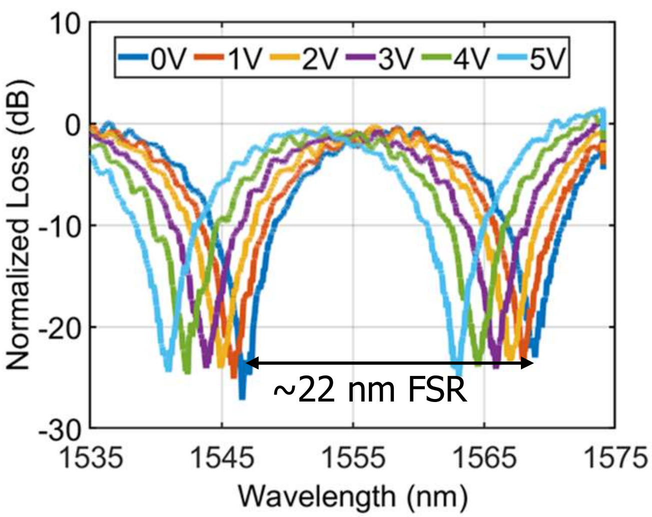


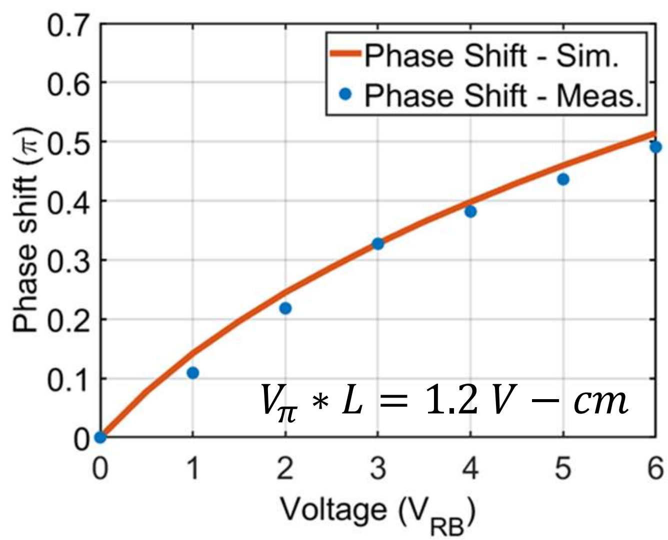


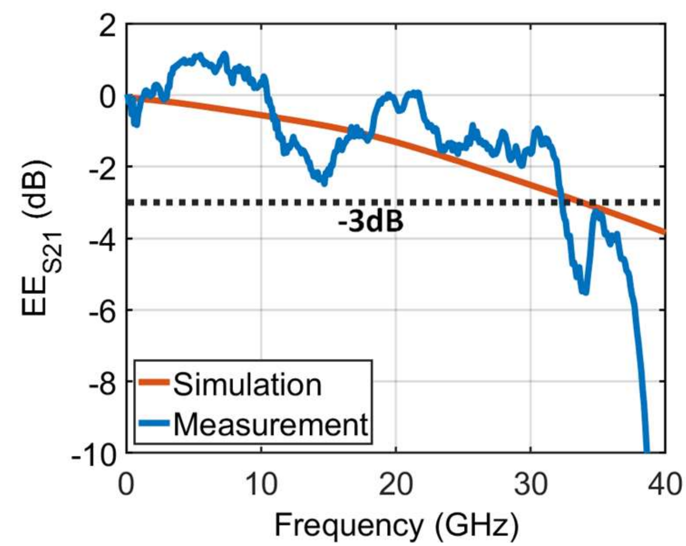
Differential MZM – Measurement











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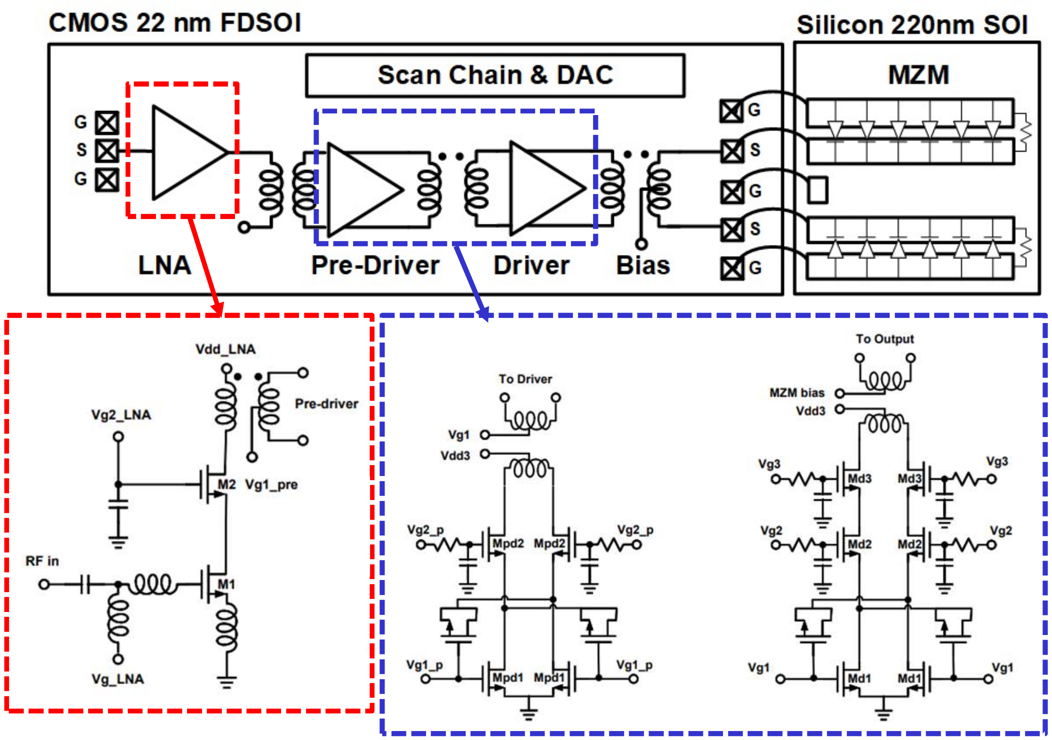
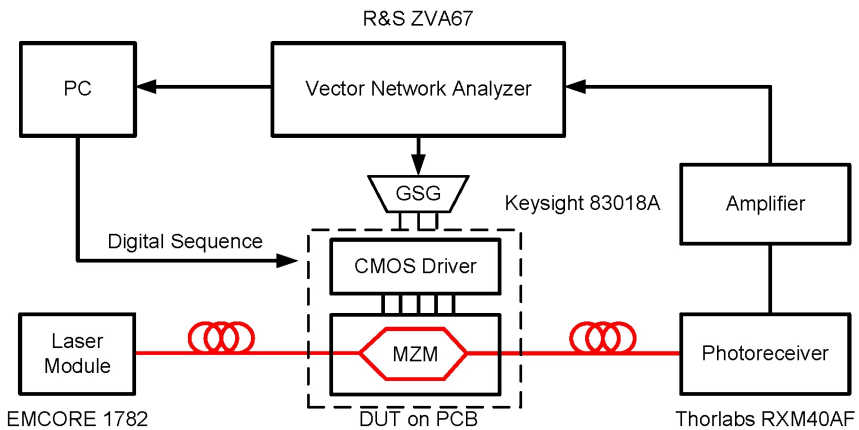
16-32 GHz driver in 22nm CMOS FD-SOI

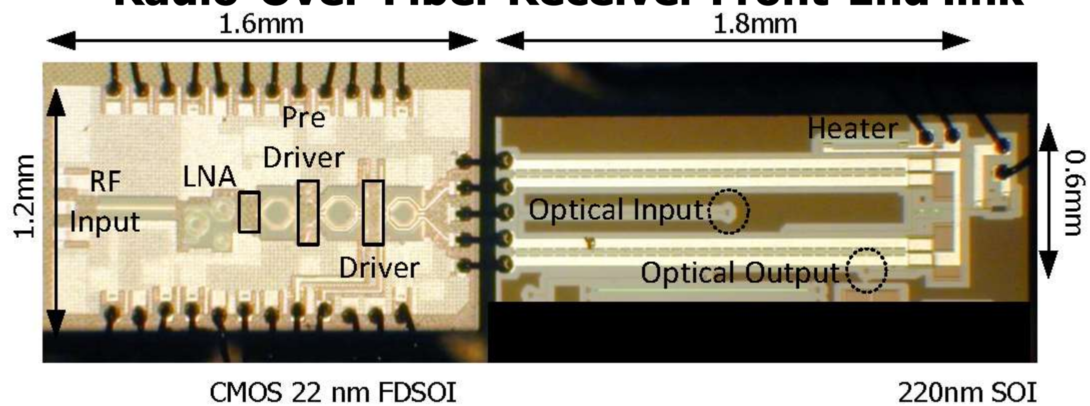
- Cascode LNA improves overall link noise performance
- Pre-Driver and Driver stages use stacked-FET structure for suppling large voltage swing to MZM
- Magnetically coupled resonator (MCR) technique is used for interstage matching at all input and output stages



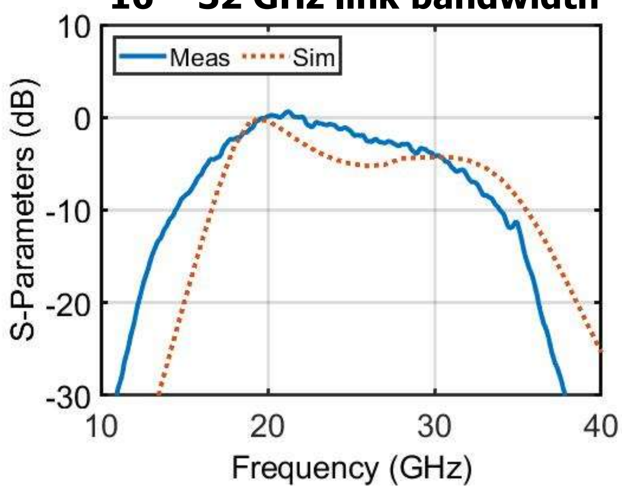
RoF Front-End Implementation & Measurements



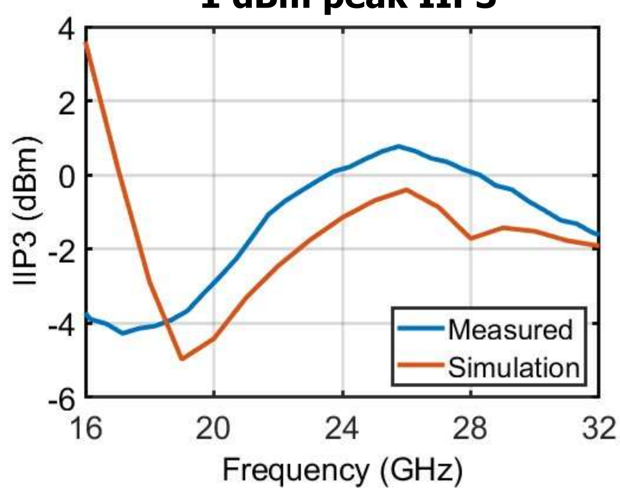
Radio-Over-Fiber Receiver Front-End link



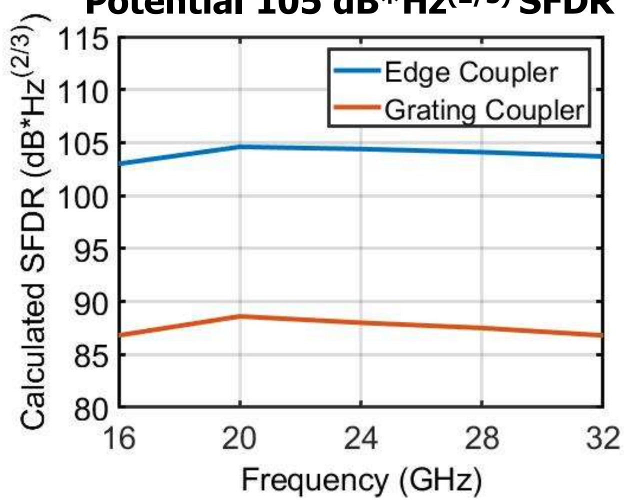
16 – 32 GHz link bandwidth



1 dBm peak IIP3



Potential 105 dB*Hz$^{(2/3)}$ SFDR



Conclusion

- Silicon Photonics is key for next-generation high-speed wireless and data links.
- A 32 GHz high-speed differential Mach-Zehnder Modulator is designed and characterized.
- Inductive slots are implemented for slow-wave transmission line design to achieve simultaneous impedance and velocity matching.
- A 16-32 GHz RoF Front-End is demonstrated with Silicon Photonic Mach-Zehnder Modulator integrated with 22nm CMOS driver with 1dB peak IIP3 and a potential 105 dB.Hz$^{2/3}$ SFDR